

Introduction to Probabilistic Machine Learning

Graphical Models: Approximate Inference

Ralf Herbrich

Overview

1. Distance Measures for Distributions
2. Approximating Marginals: Expectation Propagation
3. Approximating Normalization Constants

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*Unit 5 – Graphical Models:
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Distance Measures: α -Divergence

- **Problem.** We have a non-Gaussian normalized marginal $p(\cdot)$ and would like to approximate it by a Gaussian $q(\cdot) = \mathcal{N}(\cdot; \mu, \sigma^2)$. What is the optimal approximation?
- **Solution.** To define “optimality”, we need a distance measure between probability densities $p(\cdot)$ and $q(\cdot)$!
- **α -Divergence (Amari, 1985).** Given two probability densities $p(\cdot)$ and $q(\cdot)$ and $\alpha \in \mathbb{R}$, the α -divergence $D_\alpha[p, q]$ is defined by

$$D_\alpha[p, q] = \frac{1}{\alpha(1-\alpha)} \cdot \left(1 - \int_{-\infty}^{+\infty} \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) \, dx \right)$$

Expectation of $\left[\frac{p(x)}{q(x)} \right]^\alpha$ over $q(x)$

If $p = q$ then $\left[\frac{p(x)}{q(x)} \right]^\alpha = 1$
and the expectation is 1

- **Non-Negativity:** If $p = q$ then $D_\alpha[p, q] = 0$; otherwise $D_\alpha[p, q] > 0$
- **Asymmetry:** $D_\alpha[p, q] \neq D_\alpha[q, p]$
- **Flexibility:**
 - $\alpha > 1$ gives more weight to regions where $p(x) > q(x)$
 - $\alpha < 1$ gives more weight to regions where $q(x) > p(x)$

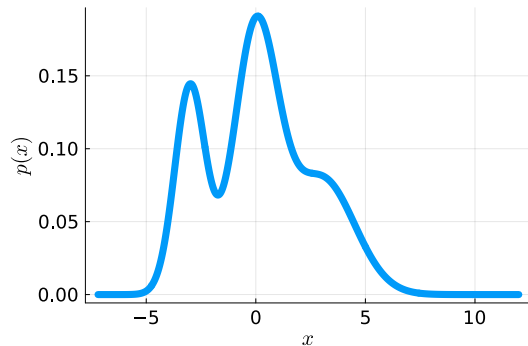


Shun'ichi Amari (甘利 俊)
(1936)

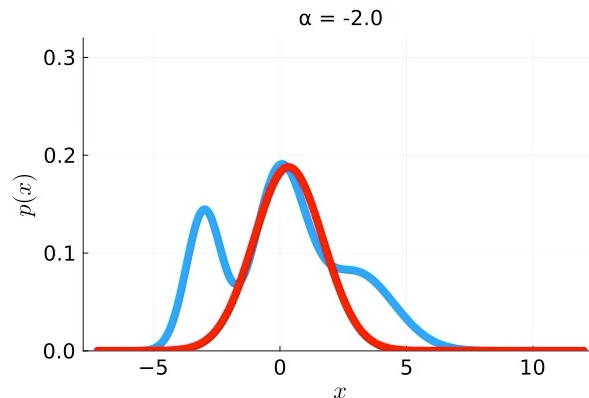
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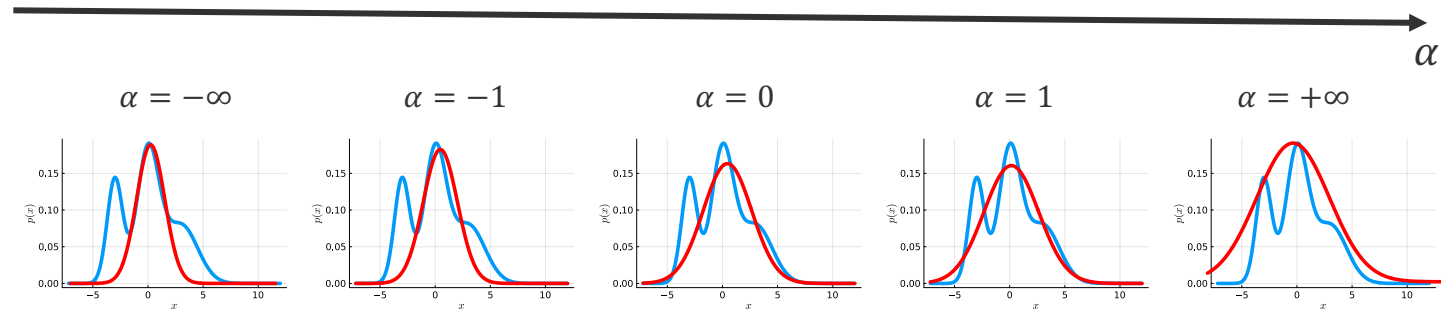
α -Divergence with a Gaussian in Pictures



zero-forcing & mode seeking



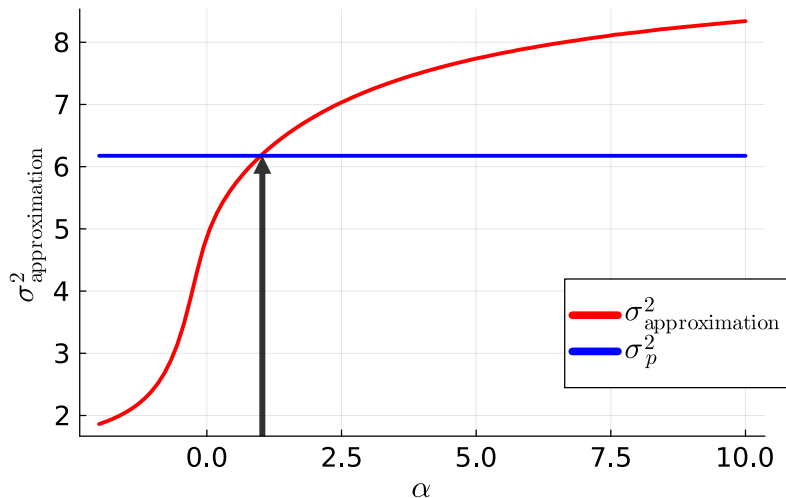
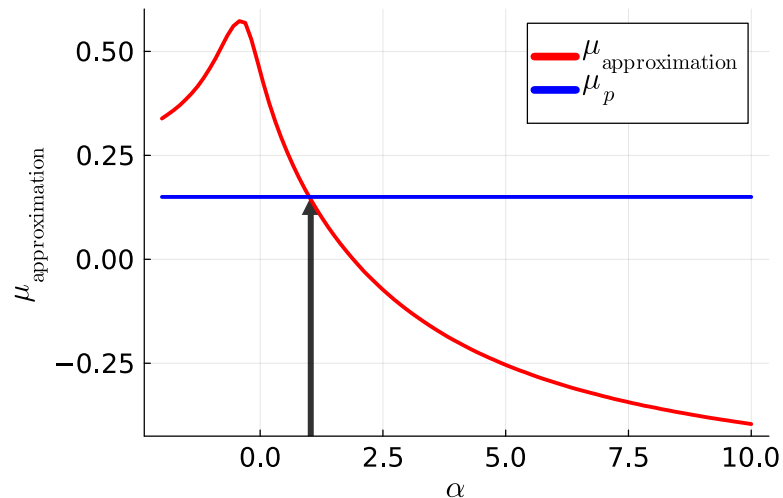
inclusive & support seeking



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α -Divergence with a Gaussian and Moment Matching



- Only for $\alpha = 1$ both the first and second moment (that is, mean and variance) are matched with that of the approximation!
- The case $\alpha = 1$ is a limit case.

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$\alpha = 1$: KL Divergence

- **Theorem (Limit $\alpha \rightarrow 1$).** Given two probability densities $p(\cdot)$ and $q(\cdot)$ the limit of the α -divergence $D_\alpha[p, q]$ for $\alpha \rightarrow 1$ is the Kullback-Leibler divergence

$$\lim_{\alpha \rightarrow 1} D_\alpha[p, q] = \text{KL}[p, q] := \int_{-\infty}^{+\infty} \log\left(\frac{p(x)}{q(x)}\right) \cdot p(x) \, dx$$

- **Proof:** Taking limits, we have

$$\begin{aligned} \lim_{\alpha \rightarrow 1} D_\alpha[p, q] &= \lim_{\alpha \rightarrow 1} \frac{1}{\alpha(1-\alpha)} \cdot \left(1 - \int_{-\infty}^{+\infty} \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) \, dx \right) \quad \text{L'Hôpital's rule!} \\ &= \lim_{\alpha \rightarrow 1} \frac{1}{1-2\alpha} \cdot \left(- \int_{-\infty}^{+\infty} \log\left(\frac{p(x)}{q(x)}\right) \cdot \left[\frac{p(x)}{q(x)} \right]^\alpha \cdot q(x) \, dx \right) \quad \text{Note that } \frac{d}{dz} b^z = \log(b) \cdot b^z \\ &= \int_{-\infty}^{+\infty} \log\left(\frac{p(x)}{q(x)}\right) \cdot p(x) \, dx \end{aligned}$$

- **Theorem (Moment Matching).** Given any distribution $p(\cdot)$ the minimizer μ^*, σ^{2*} of the KL divergence $\text{KL}[p(\cdot), \mathcal{N}(\cdot; \mu, \sigma^2)]$ to a Gaussian distribution has

$$\mu^* = E_{X \sim p(\cdot)}[X] \quad \text{and} \quad \sigma^{2*} = E_{X \sim p(\cdot)}[X^2] - (\mu^*)^2$$



Solomon Kullback
(1909 – 1994)



Richard Leibler
(1914 – 2003)

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Sum-Product Algorithm Revisited

- The key operation for factor $f(x_1, x_2, \dots, x_n)$ and variable X_1 is

$$m_{f \rightarrow X_1}(x_1) = \sum_{\{x_2\}} \cdots \sum_{\{x_n\}} f(x_1, x_2, \dots, x_n) \prod_{j=2}^n m_{X_j \rightarrow f}(x_j)$$

If all $m_{X_j \rightarrow f}(x_j)$ are Gaussian, the result **might not be** Gaussian!

- Based on outgoing messages, we can compute both non-normalized marginals $p_X(\cdot)$ and $m_{X \rightarrow f}(\cdot)$

$$p_X(x) = \prod_{f \in \text{ne}(X)} m_{f \rightarrow X}(x) \quad m_{X \rightarrow f}(x) = \frac{p_X(x)}{m_{f \rightarrow X}(x)}$$

If all $m_{X_j \rightarrow f}(x_j)$ are Gaussian, the result **must be** Gaussian!

Idea:

- We approximate all outgoing messages $m_{f \rightarrow X}(\cdot)$ by a Gaussian $\hat{m}_{f \rightarrow X}(\cdot) = \mathcal{N}(\cdot; \mu, \sigma^2)$
- We measure the approximation quality in the normalized marginal, **not** the outgoing message

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$$\hat{p}(\cdot) = \arg \min_{\mu, \sigma^2} \text{KL} \left[\frac{m_{f \rightarrow X}(\cdot) \cdot \hat{m}_{X \rightarrow f}(\cdot)}{\int_{-\infty}^{+\infty} m_{f \rightarrow X}(\tilde{x}) \cdot \hat{m}_{X \rightarrow f}(\tilde{x}) d\tilde{x}}, \frac{\mathcal{N}(\cdot; \mu, \sigma^2) \cdot \hat{m}_{X \rightarrow f}(\cdot)}{\int_{-\infty}^{+\infty} \mathcal{N}(\tilde{x}; \mu, \sigma^2) \cdot \hat{m}_{X \rightarrow f}(\tilde{x}) d\tilde{x}} \right]$$

True normalized marginal with
approximate incoming message

Approximate marginal with
approximate incoming message

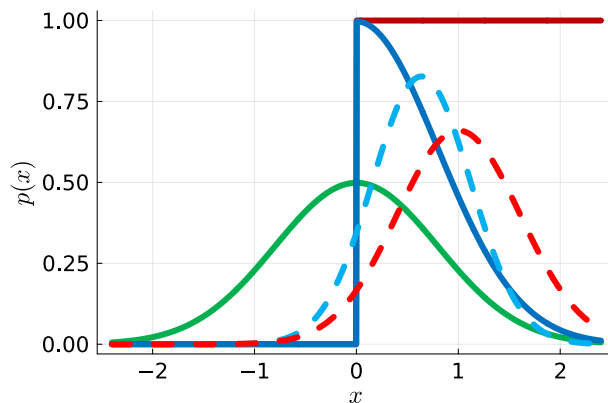
Approximate Message Passing: Example

$$f(x) = \mathbb{I}(x > 0)$$

$$\hat{m}_{X \rightarrow f}(x) \propto \frac{\hat{p}_X(x)}{\hat{m}_{f \rightarrow X}(x)} \longrightarrow p_X(x) \propto f(x) \cdot \hat{m}_{X \rightarrow f}(x)$$

$$\hat{m}_{f \rightarrow X}(x) \propto \frac{\hat{p}_X(x)}{\hat{m}_{X \rightarrow f}(x)}$$

$$\hat{p}_X(x) = \mathcal{N}(x; E_{X \sim p_X}[X], \text{var}_{X \sim p_X}[X])$$



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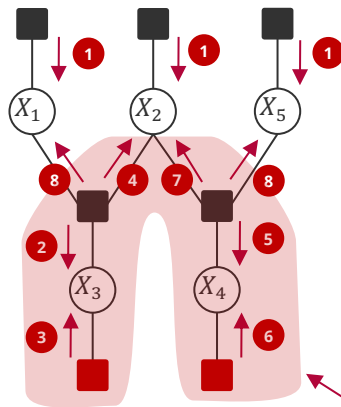
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Expectation Propagation

- **Idea:** If we have factors in the factor graph that require approximate messages, we keep iterating on the whole path between them until convergence minimizing $KL(p(\cdot) | \mathcal{N}(\cdot; \mu, \sigma^2))$ locally for the affected marginals of the approximate factor.
- **Theorem (Minka, 2003):** *The approximate message passing algorithm using the Kullback-Leibler convergence will always converge if the approximating distribution is in the exponential family!*



Tom Minka



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Doubly-Truncated Gaussians

- **Doubly-Truncated Gaussian.** Given $l, u \in \mathbb{R}$, $\mu \in \mathbb{R}$ and $\sigma^2 \in \mathbb{R}^+$, a random variable X has a doubly-truncated Gaussian distribution if

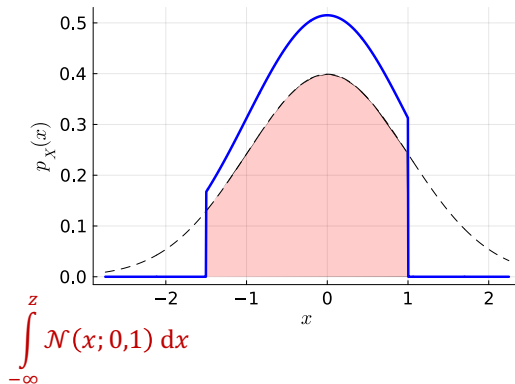
$$p_X(x) \propto \mathbb{I}(l < x < u) \cdot \mathcal{N}(x; \mu, \sigma^2)$$

- **Moments of Doubly-Truncated Gaussian.** Given a random variable X that has a doubly-truncated Gaussian distribution and $t_a := a/\sigma$, we know

$$E[X^0] = \Phi(t_{u-\mu}) - \Phi(t_{l-\mu})$$

$$E[X^1] = \mu + \sigma \cdot \frac{\mathcal{N}(t_{l-\mu}) - \mathcal{N}(t_{u-\mu})}{\Phi(t_{u-\mu}) - \Phi(t_{l-\mu})}$$

$$E[X^2] = \mu^2 + \sigma^2 \cdot \left[1 - \frac{t_{u+\mu} \cdot \mathcal{N}(t_{u-\mu}) - t_{l+\mu} \cdot \mathcal{N}(t_{l-\mu})}{\Phi(t_{u-\mu}) - \Phi(t_{l-\mu})} \right]$$



Additive correction that goes to zero as $u \rightarrow \infty$ and $l \rightarrow -\infty$

Multiplicative correction that goes to one as $u \rightarrow \infty$ and $l \rightarrow -\infty$

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Doubly-Truncated Gaussians (ctd)

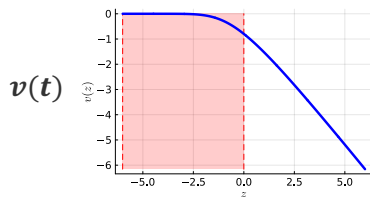
- Using the variance decomposition theorem we see

$$E[X] = \mu + \sigma \cdot v_{l,u} \left(\frac{\mu}{\sigma} \right) \quad \text{var}[X] = \sigma^2 \cdot \left[1 - w_{l,u} \left(\frac{\mu}{\sigma} \right) \right]$$

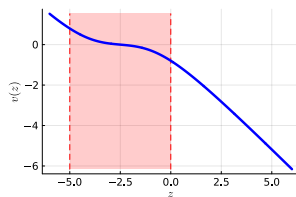
$$v_{l,u}(z) := \frac{\mathcal{N}(l-z) - \mathcal{N}(u-z)}{\Phi(u-z) - \Phi(l-z)}$$

$$w_{l,u}(z) := \frac{(u+z) \cdot \mathcal{N}(u-z) - (l+z) \cdot \mathcal{N}(l-z)}{\Phi(u-z) - \Phi(l-z)} + v_{l,u}(z) \cdot [2z + v_{l,u}(z)]$$

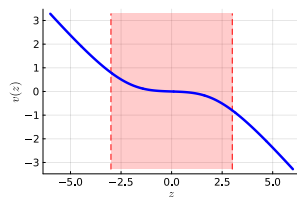
$x < 0$



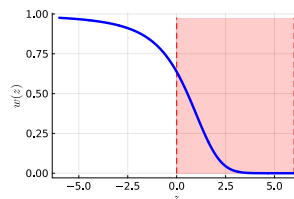
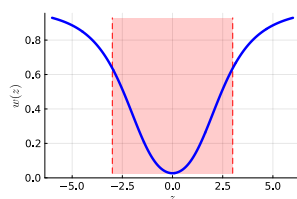
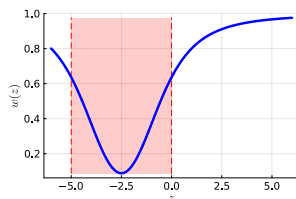
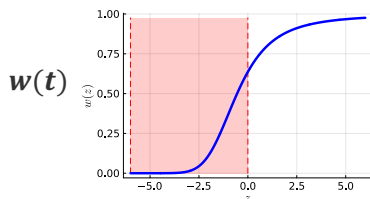
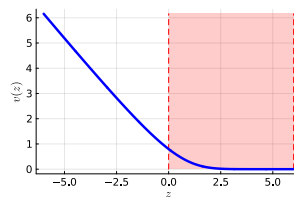
$-5 \leq x < 0$



$-3 \leq x < 3$



$0 \leq x$



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Normalization Constant

- **Normalization Constant:** Given a factor graph with factors $f_1, \dots, f_m$, each over a subset of n variables $X_1, X_2, \dots, X_n$, the normalization constant Z is defined as the sum over all variables

$$Z = \sum_{\{x_1\}} \cdots \sum_{\{x_n\}} f_1(\mathbf{x}_{\text{ne}(f_1)}) \cdot f_2(\mathbf{x}_{\text{ne}(f_2)}) \cdots f_m(\mathbf{x}_{\text{ne}(f_m)})$$

- **Inference in Factor Graphs:** In order to learn from data D we

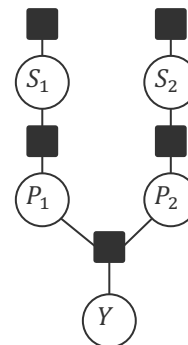
1. **Modelling:** Formulate a joint model $p(\boldsymbol{\theta}, D)$ of parameters $\boldsymbol{\theta} = \theta_1, \dots, \theta_n$ and data D

$$Z = \sum_{\{\theta_1\}} \cdots \sum_{\{\theta_n\}} \sum_{\{D\}} p(\boldsymbol{\theta}, D) = 1$$

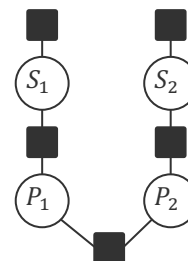
2. **Conditioning:** Remove the variables that represent data D from the factor graph

$$Z = \sum_{\{\theta_1\}} \cdots \sum_{\{\theta_n\}} p(\boldsymbol{\theta}, D) = p(D) \quad \leftarrow \text{Law of total probability}$$

- The normalization constant is the probability of the data D and measures how good our probabilistic model explains the observed training set D (*model evidence*)!



$$p(S_1, S_2, P_1, P_2, Y)$$



$$p(S_1, S_2, P_1, P_2, Y = 1)$$

Normalization Constant via Message Passing

- The non-normalized marginal $p_{X_j}(\cdot)$ is defined as

$$p_{X_j}(x_j) = \sum_{\{x_1\}} \cdots \sum_{\{x_{j-1}\}} \sum_{\{x_{j+1}\}} \cdots \sum_{\{x_n\}} f_1(\mathbf{x}_{\text{ne}(f_1)}) \cdot f_2(\mathbf{x}_{\text{ne}(f_2)}) \cdots f_m(\mathbf{x}_{\text{ne}(f_m)})$$

$$\prod_{i \in \text{ne}(X_j)} m_{f_i \rightarrow X_j}(x_j)$$

Variable node root

$$Z = \sum_{\{x_j\}} \prod_{i \in \text{ne}(X_j)} m_{f_i \rightarrow X_j}(x_j)$$

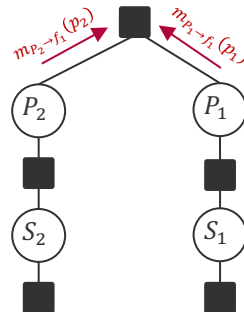
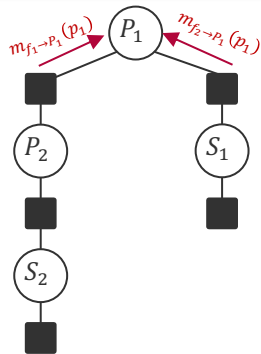
$$Z = \sum_{\{x_j\}} p_{X_j}(x_j)$$

Factor node root

$$Z = \sum_{\{\mathbf{x}_{\text{ne}(f_i)}\}} f(\mathbf{x}_{\text{ne}(f_i)}) \prod_{k \in \text{ne}(f_i)} m_{X_k \rightarrow f_i}(x_k)$$

$$m_{f_i \rightarrow X_j}(x_j) \cdot m_{X_j \rightarrow f_i}(x_j)$$

$$\sum_{\{\mathbf{x}_{\text{ne}(f_i) \setminus \{j\}}\}} f(\mathbf{x}_{\text{ne}(f_i)}) \prod_{k \in \text{ne}(f_i) \setminus \{j\}} m_{X_k \rightarrow f_i}(x_k)$$



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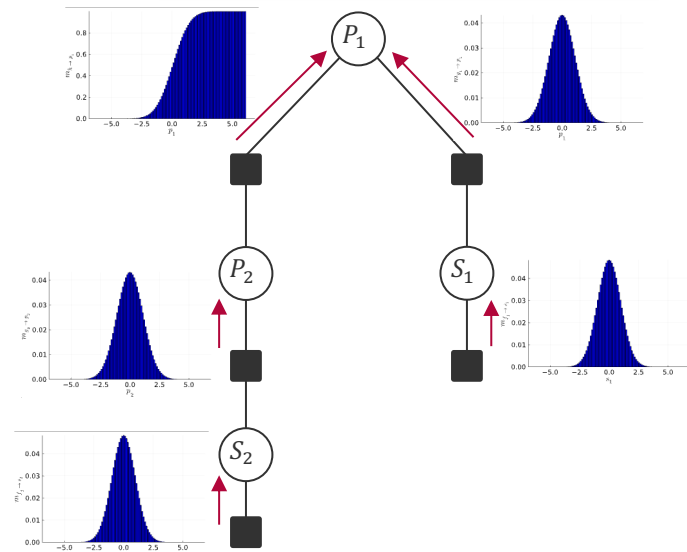
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Normalization Constant via Message Passing: Example

$$\mathcal{N}(s_1; \mu_1, \sigma_1^2) \cdot \mathcal{N}(s_2; \mu_2, \sigma_2^2) \cdot \mathcal{N}(p_1; s_1, \beta^2) \cdot \mathcal{N}(p_2; s_2, \beta^2) \cdot \mathbb{I}(y(p_1 - p_2) > 0)$$

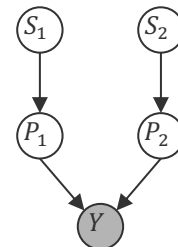
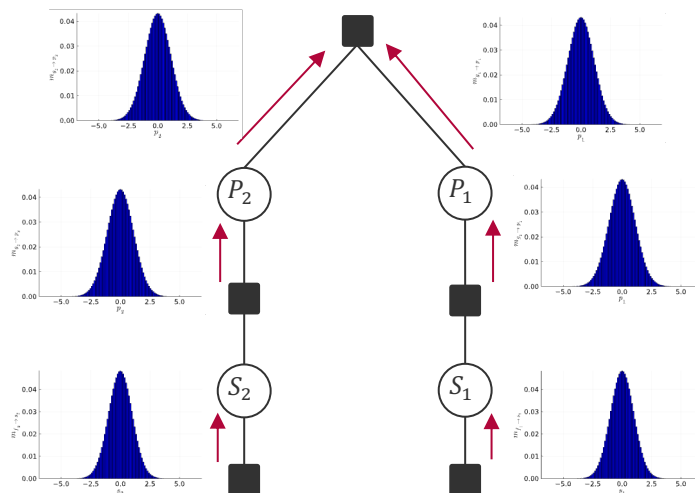
Variable node root

$$Z = \sum_{\{x_j\}} \prod_{i \in \text{ne}(X_j)} m_{f_i \rightarrow X_j}(x_j)$$



Factor node root

$$Z = \sum_{\{x_{\text{ne}(f_i)}\}} f(x_{\text{ne}(f_i)}) \prod_{k \in \text{ne}(f_i)} m_{X_k \rightarrow f_i}(x_k)$$



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Normalization Constant via Approximate Message Passing

- **Challenge:** Computing the normalization constant Z requires
 1. Tracking the normalization constants for all messages
 2. Choosing a variable X_j or factor f_i as a root of the factor tree
- **Observation:** Tracking the normalization constant for all messages is not always possible when approximating messages!
- **Theorem:** Given a factor tree with renormalized messages $\tilde{m}_{X_j \rightarrow f_i}(\cdot) = \beta_{j,i} \cdot m_{X_j \rightarrow f_i}(\cdot)$ and $\tilde{m}_{f_i \rightarrow X_j}(\cdot) = \alpha_{i,j} \cdot m_{f_i \rightarrow X_j}(\cdot)$ the normalization constant Z is

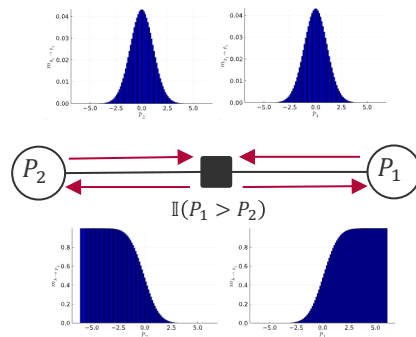
$$Z = \left(\prod_{i=1}^m Z_{f_i} \right) \cdot \left(\prod_{j=1}^n Z_{X_j} \right)$$

Factor Normalization

Variable Normalization

$$Z_{f_i} = \frac{\sum_{\{x_{\text{ne}(f_i)}\}} f(x_{\text{ne}(f_i)}) \prod_{k \in \text{ne}(f_i)} \tilde{m}_{X_k \rightarrow f_i}(x_k)}{\sum_{\{x_{\text{ne}(f_i)}\}} \prod_{k \in \text{ne}(f_i)} \tilde{m}_{f_i \rightarrow X_k}(x_k) \cdot \tilde{m}_{X_k \rightarrow f_i}(x_k)}$$

$$Z_{X_j} = \sum_{\{x_j\}} \prod_{i \in \text{ne}(X_j)} \tilde{m}_{f_i \rightarrow X_j}(x_j)$$



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Proof of Normalization Theorem

- Simplifying the variable normalization Z_{X_j} and factor normalization Z_{f_i}

Variable normalization

$$\begin{aligned}
 Z_{X_j} &= \sum_{\{x_j\}} \prod_{i \in \text{ne}(X_j)} \tilde{m}_{f_i \rightarrow X_j}(x_j) \quad \leftarrow \alpha_{i,j} \cdot m_{f_i \rightarrow X_j}(x_j) \\
 &= \left(\prod_{i \in \text{ne}(X_j)} \alpha_{i,j} \right) \cdot \left(\sum_{\{x_j\}} \prod_{i \in \text{ne}(X_j)} m_{f_i \rightarrow X_j}(x_j) \right) \quad \leftarrow Z \\
 &= \left(\prod_{i \in \text{ne}(X_j)} \alpha_{i,j} \right) \cdot Z
 \end{aligned}$$

Factor normalization

$$\begin{aligned}
 Z_{f_i} &= \frac{\sum_{\{x_{\text{ne}(f_i)}\}} f(x_{\text{ne}(f_i)}) \prod_{k \in \text{ne}(f_i)} \tilde{m}_{X_k \rightarrow f_i}(x_k) \quad \leftarrow \beta_{k,i} \cdot m_{X_k \rightarrow f_i}(x_k)}{\sum_{\{x_{\text{ne}(f_i)}\}} \prod_{k \in \text{ne}(f_i)} \tilde{m}_{f_i \rightarrow X_k}(x_k) \cdot \tilde{m}_{X_k \rightarrow f_i}(x_k)} \\
 &= \frac{(\prod_{k \in \text{ne}(f_i)} \beta_{k,i}) \cdot \left(\sum_{\{x_{\text{ne}(f_i)}\}} f(x_{\text{ne}(f_i)}) \prod_{k \in \text{ne}(f_i)} m_{X_k \rightarrow f_i}(x_k) \right) \quad \leftarrow Z}{(\prod_{k \in \text{ne}(f_i)} \beta_{k,i}) \cdot \left(\prod_{k \in \text{ne}(f_i)} \alpha_{i,k} \right) \cdot \left(\sum_{\{x_{\text{ne}(f_i)}\}} \prod_{k \in \text{ne}(f_i)} m_{f_i \rightarrow X_k}(x_k) \cdot m_{X_k \rightarrow f_i}(x_k) \right)} \\
 &= \frac{Z}{\left(\prod_{k \in \text{ne}(f_i)} \alpha_{i,k} \right) \cdot Z^{|\text{ne}(f_i)|}}
 \end{aligned}$$

$\prod_{k \in \text{ne}(f_i)} \sum_{\{x_k\}} m_{f_i \rightarrow X_k}(x_k) \cdot m_{X_k \rightarrow f_i}(x_k)$

$$\left(\prod_{j=1}^n Z_{X_j} \right) \cdot \left(\prod_{i=1}^m Z_{f_i} \right) = \left(\prod_{j=1}^n \prod_{i \in \text{ne}(X_j)} \alpha_{i,j} \right) \cdot Z^n \cdot \frac{Z^m}{\left(\prod_{i=1}^m \prod_{k \in \text{ne}(f_i)} \alpha_{i,k} \right) \cdot \prod_{i=1}^m Z^{|\text{ne}(f_i)|}}$$

identical

Z^{n+m-1} because the factor graph is a tree with $n + m$ nodes!

1. Distance Measures for Distributions

- α -divergences are a general class of distance measures between distributions
- For $\alpha = 1$, the α -divergence becomes the Kullback-Leibler divergence where the minimizer for Gaussian approximating distributions matches mean and variance
- Minimizers of α -divergences range from mode-seeking to support-seeking

2. Approximate Message Passing and Expectation Propagation

- Approximations will always be done on the marginals, **not** the messages
- When the Kullback-Leibler divergence is used as distance, all moments get preserved
- In case of doubly-truncated Gaussians, the moments are closed form

3. Approximating Normalization Constants

- If factor graphs represent joint probabilities of data and parameters with data variables only, the normalization constant equals the probability of data (under the model)
- If messages can be explicitly represented and normalized, efficient updates can be done starting at any factor or variable
- There is a general distributed algorithm for approximating the normalization constant

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See you next week!